SSC MTS 2024 question paper with answers

22 October 2024 · Shift 1

76 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 14 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 12 questions

Q.1

During the evening patrolling, a policeman noticed a thief snatching a lady's mobile phone at a distance of 200 m. The thief starts running at a speed of 15 km/hr. If the policeman is chasing him at a speed of 16 km/hr, then the time taken to catch the thief is:

(A) 30 minutes
(B) 18 minutes
(C) 12 minutes
(D) 24 minutes

Q.2

There are two types of dal. Type 1 dal costs ₹100 per kg and type 2 dal costs ₹120 per kg, respectively. If both the dal are mixed in the ratio 2 : 3, find the cost of the mixed dal per kg.

(A) ₹110
(B) ₹112
(C) ₹115
(D) ₹114

Q.3

When 80% of a number is added to 80, the result is the same number. Find the number.

(A) 320
(B) 400
(C) 420
(D) 380

Q.4

Find the HCF of the numbers 216, 360 and 504.

(A) 252
(B) 72
(C) 108
(D) 180

Q.5

A borrows ₹7,500 for 4 years at 8% p.a. simple interest. B borrows the same amount at simple interest of 9% for the first year, 10% for the second year and 6% for the next two years. What is simple interest paid by A and B in total (in ₹)?

(A) 4,275
(B) 4,545
(C) 4,175
(D) 4,725

Q.6

Aditya bought a mobile for ₹21,500 and sold it for ₹18,500. Find the loss percentage (up to two decimal places) in the complete process.

(A) 12.38%
(B) 13.95%
(C) 10.69%
(D) 11.25%

Q.8

In a box, Refrigerator covers and Washing Machine covers are in the ratio of 7 : 11. If the Washing Machine covers are 660, how many Refrigerator covers are there?

(A) 420
(B) 400
(C) 410
(D) 430

Q.10

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 9% per year rate of interest for 4 years.

(A) 1080
(B) 1040
(C) 1020
(D) 1120

Q.11

P, Q and R start running around a circular track and complete one round in 18, 22 and 30 seconds, respectively. In how much time will they meet again at the starting point?

(A) 16 min 30 sec
(B) 16 min 50 sec
(C) 11 min 30 sec
(D) 11 min 50 sec

Q.13

The average of runs scored by a player in 22 innings is 40 runs. How many runs should he score in the 23rd innings so that his average is increased by 6 runs?

(A) 188 runs
(B) 178 runs
(C) 158 runs
(D) 148 runs

Q.14

What sum (in ₹) will earn a simple interest of ₹240 in 4 years at 8% per year rate of interest?

(A) 375
(B) 750
(C) 1500
(D) 1000

Q.20

The average of 42 numbers is 25 and that of other 28 numbers is 40. What is the average of all the numbers?

(A) 69
(B) 31
(C) 70
(D) 30

Reasoning Ability and Problem Solving 15 questions

Q.2

This question has two statements followed by two conclusions numbered I and II. You have to take the given statements to be true even if they seem to be at variance from commonly known facts. You have to decide which of the given conclusions logically follow(s) from the given statements. Statements: All jugs are taps. Some taps are hats. Conclusions: (I) Some taps are jugs. (II)All hats are jugs.

(A) Both conclusions (I) and (II) follow
(B) Neither conclusion (I) nor (II) follows
(C) Only conclusion (I) follows
(D) Only conclusion (II) follows

Q.3

In a certain code language, ‘MELT’ is coded as ‘6284’ and ‘TUNE’ is coded as ‘9758’. What is the code for ‘T’ in that language?

(A) 4
(B) 5
(C) 9
(D) 8

Q.4

In a certain code language, ‘SANE’ is coded as ‘5168’ and ‘GOAT’ is coded as ‘6437’.What is the code for ‘A’ in the given code language?

(A) 5
(B) 6
(C) 3
(D) 1

Q.6

What should come in place ofthe question mark (?) in the given series? 5 ,15, 45, 135 ,405 ,?

(A) 1125
(B) 1215
(C) 1315
(D) 1255

Q.7

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 - Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (5, 40, 7) (12, 65, 5)

(A) (2, 10, 3)
(B) (6, 24, 2)
(C) (3, 14, 3)
(D) (9, 41,5)

Q.9

I, J, K, L, M, N and O are sitting around a circular table facing the centre (but not necessarily in the same order). J sits fourth to the right of I. K sits to the immediate right of J. L sits to the immediate right of I. N sits fourth to the left of M. M is not an immediate neighbour of K. How many people sit between O and N when counted from the right of O?

(A) 4
(B) 2
(C) 5
(D) 3

Q.10

DG 25 is related to FH 21 in a certain way. In the same way, GI 19 is related to IJ 15. To which of the following is MI 24 related, following the same logic?

(A) OJ 20
(B) OK 20
(C) PJ 20
(D) PO 20

Q.11

What will come in place of the question mark (?) in the following equation if ‘+’ and ‘×’ are interchanged and ‘–’ and ‘÷’ are interchanged? 14 ÷ 12 – 2 + 2 × 16 = ?

(A) 18
(B) 24
(C) 42
(D) 20

Q.12

Find the missing term in the below series: B2, C3, E5, G7, __, M13, Q17.

(A) I10
(B) H7
(C) K11
(D) J10

Q.13

UKCT is related to QGYP in a certain way based on the English alphabetical order. In the same way, LPBR is related to HLXN. To which of the following is DGYJ related, following the same logic?

(A) YCSE
(B) ZCUF
(C) XBUF
(D) ZBSF

Q.14

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: Some potatoes are tomatoes. No tomato is an onion. Conclusions: (I) Some tomatoes may be potatoes. (II) No potato is an onion.

(A) Only conclusion I follows
(B) Both conclusions I and II follow
(C) Only conclusion II follows
(D) Neither conclusion I nor II follows

Q.15

Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) (8, 24, 12) (16, 20, 5)

(A) (17, 75, 18)
(B) (21, 88, 16)
(C) (13, 24, 8)
(D) (19, 57, 12)

Q.16

Seven people A, B, C, D, E, F and G are sitting around a circular table, facing the centre (but not necessarily in the same order). Exactly three people are sitting between A and B when counted from the right of A. Exactly two people are sitting between D and G when counted from the left of G. B is not an immediate neighbour of G. D is sitting to the immediate left of B. Exactly three people are sitting between E and F when counted from the right of E. B and E are not immediate neighbours. How many people are sitting between E and G when counted from the right of E?

(A) 4
(B) 3
(C) 2
(D) 1

Q.18

In a certain code language, ‘COPY’ is coded as ‘2465’ and ‘PLAN’ is coded as ‘3759’. What is the code for ‘P’ in the given code language?

(A) 5
(B) 3
(C) 6
(D) 7

Q.20

What will come in place of the question mark (?) in the following equation if ‘+’ and ‘–’ are interchanged and ‘×’ and ‘÷’ are interchanged? $3 - 2 \div 8 \times 4 + 2 = ?$

(A) 8
(B) 14
(C) 17
(D) 5

General Awareness 24 questions

Q.1

Identify the correct statement with reference to the Padma awards.

(A) The names of the awardees are never published in the Gazette of India.
(B) The total number of Padma awards to be given in a year should not more than 150.
(C) The Padma Awards are given annually.
(D) The Padma award can be used as a suffix or prefix to the awardees’ name

Q.2
Which of the following is NOT an element of weather?

(A) Humidity
(B) Plants
(C) Temperature
(D) Rainfall

Q.3
In which of the following years India hosted the Women's Asia Cup ODI Cricket?

(A) 2006
(B) 2005
(C) 2007
(D) 2004

Q.4
The famous English novel ‘Em and the Big Hoom’ is written by:

(A) Jerry Pinto
(B) Esther David
(C) Amitav Ghosh
(D) Shashi Tharoor

Q.5
The Prime Minister in India is _____.

(A) the real executive
(B) the Supreme Commander of Armed Forces
(C) a nominal executive
(D) the head of the judiciary

Q.6
Which of the following is NOT a component of revenue receipt?

(A) Fees paid for the registration of property
(B) Amount borrowed from World Bank
(C) GST
(D) Grant received by UNESCO

Q.7
Organisation of agriculture and animal husbandry is included in which of the following Articles of the Indian Constitution?

(A) Article 46
(B) Article 47
(C) Article 48
(D) Article 45

Q.8
0.25 g of an organic compound, on treatment with fuming nitric acid followed by $AgNO_3$, gave 0.22 g of AgBr. The percentage of bromine in the compound is _____.

(A) 37.45%
(B) 34.04%
(C) 50.00%
(D) 88.00%

Q.9
In Union Budget 2023-24, annual production of 5 MMT under Green Hydrogen Mission is to be targeted by the year _____ to facilitate transition of the economy to low carbon intensity.

(A) 2032
(B) 2023
(C) 2027
(D) 2030

Q.10
According to the Census of India-2011, what is the share of workforce engaged in the agriculture sector?

(A) 65.3%
(B) 48.9%
(C) 45.8%
(D) 54.6%

Q.11
Alarmel Valli was associated with which of the following dances?

(A) Manipuri
(B) Bharatanatyam
(C) Odissi
(D) Kathak

Q.12
When protein-rich gelatinous egg white is heated, it turns to solid white due to:

(A) Protein Denaturation
(B) Fermentation
(C) Emulsification
(D) Gluten Formation

Q.13
Which of the following is an example of the first garden-tomb on the Indian subcontinent?

(A) Safdarjung's tomb
(B) Tomb of Itimad-ud-Daulah
(C) Tomb of Tansen
(D) Humayun’s tomb

Q.14
In which of the following Indian states is the M.A. Chidambaram Stadium located?

(A) Karnataka
(B) West Bengal
(C) Tamil Nadu
(D) Kerala

Q.15
In April 2023, who was appointed as the chairman of the National Technical Research Organisation (NTRO) by the centre?

(A) Arvind Kumar
(B) Ajit Doval
(C) Anil Dhasmana
(D) Arun Sinha

Q.17
In which of the following states are the Nor Westers known as ‘Bardoisila’?

(A) Goa
(B) Bihar
(C) Assam
(D) Odisha

Q.18
Rani ki Vav is a UNESCO world heritage monument, situated in which of the following states?

(A) Maharashtra
(B) Karnataka
(C) Haryana
(D) Gujarat

Q.19
According to the Census 2011, what was the annual population growth rate of India (in percentage)?

(A) 2.16%
(B) 2.22%
(C) 1.97%
(D) 1.64%

Q.20
Usha Barle, who won Padma Shri 2023, is popularly associated with which genre of music?

(A) Panihari
(B) Kajari
(C) Pandwani
(D) Lavani

Q.21
Which of the following statements is/are correct? A)Balshastri Jambhekar started the first Anglo- Maratha newspaper - ‘Durpun’, in 1832. B)He is remembered as the father of Marathi journalism. C)Balshastri Jambhekar published the first Marathi monthly magazine - ‘Digdurshun’ in 1840.

(A) Only B
(B) Both A and B
(C) Only A
(D) A, B and C

Q.22
Which of the following organisms is used in bread and alcohol industries?

(A) Penicillium
(B) Nitrobacter
(C) Clostridium
(D) Yeast

Q.23
What is the collection of Vedic hymns or ‘mantras’ called?

(A) Samhita
(B) Smriti
(C) Upanishad
(D) Shruti

Q.24
The Indian Forest Service was created under the All India Service Act, 1951 in _____.

(A) 1957
(B) 1966
(C) 1954
(D) 1060

Q.25
Bathukamma festival is celebrated in which of the following states?

(A) Manipur
(B) Chhattisgarh
(C) Telangana
(D) Uttar Pradesh

English Language and Comprehension 25 questions

Q.1
Select the most appropriate ANTONYM of the given word. Tranquil

(A) Serene
(B) Composed
(C) Unquiet
(D) Shabby

Q.2
Select the most appropriate option to fill in the blank. I couldn't believe how _____ the math problem was; it took me hours to solve it.

(A) complicated
(B) simply
(C) easily
(D) good

Q.3
In the given sentence, one of the four underlined words is incorrectly spelt. Identify the INCORRECTLY spelt word. She “dabbled” in “drawing” and painting, but “scalpture” was her “favourite” medium.

(A) drawing
(B) scalpture
(C) dabbled
(D) favourite

Q.4
Select the most appropriate synonym of the given word. Commence

(A) Exert
(B) Escort
(C) Start
(D) Subvert

Q.5
Read the following sentence and select its future tense form from the options given below: Sarah and Fidha entered into an agreement to share the resources that are currently accessible.

(A) Sarah and Fidha are entering into an agreement to share the resources that are currently accessible.
(B) Sarah and Fidha had entered into an agreement to share the resources that are currently accessible.
(C) Sarah and Fidha were entered into an agreement to share the resources that are currently accessible.
(D) Sarah and Fidha will enter into an agreement to share the resources that are currently accessible.

Q.6
Select the correct spelling to fill in the blank. Wedding is an auspicious _____ for all.

(A) occation
(B) occasion
(C) occassion
(D) ocassion

Q.7

Select the correct indefinite article to fill in the blank. My neighbour is _____ passionate photographer.

(A) one
(B) a
(C) an
(D) the

Q.8

Select the most appropriate option to fill in the blank. Confession is good for the soul _____ after the soul has been claimed.

(A) since
(B) even
(C) though
(D) still

Q.9

Identify the word with a spelling error in the following sentence. Select the option that correctly rectifies the error. In Shakespeare's play 'Hamlet', Hamlet's flaw is defined as procrustineition.

(A) procrastination
(B) procristineition
(C) procrustination
(D) pracrostination

Q.10

Parts of the following sentence have been given as options. Select the option that contains a grammatical error. Ms. Malathi is old much to join a college now.

(A) a college now
(B) Ms. Malathi is
(C) to join
(D) old much

Q.11

Select the sentence with the most appropriate comparative degree.

(A) She is the most talented musician in the orchestra.
(B) She is more talented musician in the orchestra.
(C) She is the talent of all musicians in the orchestra.
(D) She is the talentedest musician in the orchestra.

Q.12

Select the most appropriate option to fill in the blank. She was a _____, _____ _____astronaut.

(A) smart, tall, Indian
(B) tall, smart, Indian
(C) tall, Indian, smart
(D) smart, Indian, tall

Q.13

Select the most appropriate synonym for the given word. Species

(A) Multiply
(B) Detail
(C) Whole
(D) Type

Q.14

Select the most appropriate ANTONYM of the bracketed word in the sentence given below. Surrounded by lush greenery, the house stood _____ (submissively) atop a hill.

(A) Proudly
(B) Civilly
(C) Silently
(D) Magically

Q.15

Select the most appropriate option to fill in the blanks. When Mr. Bhanot was told a few years back that he needed to lose _____ or face health problems, he transformed his life with the _____ of a personal trainer.

(A) wait; assistance
(B) patience; guidance
(C) money; instructions
(D) weight; help

Q.16

Based on the situation in the sentence, choose the most appropriate idiom to fill in the blank. You cannot _____ before replicating your successes and achievements multiple times; otherwise, people will deem your success a stroke of dumb luck.

(A) rest on your laurels
(B) play devil's advocate
(C) sit on the fence
(D) cut to the chase

Q.17

Select the most appropriate option to substitute the underlined segment in the given sentence. Shebna decided to take “a break in social media”, but she found it difficult to disconnect completely.

(A) a break down social media
(B) a break from social media
(C) a break of social media
(D) a break up social media

Q.18

Select the most appropriate ANTONYM of the given word. Callous

(A) Tender
(B) Augment
(C) Desolate
(D) Beguile

Q.19

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. Mary is the “quieter” student of the class.

(A) more quiet
(B) quietest
(C) most quiet
(D) quiet

Q.20

Select the most appropriate meaning of the given idiom. Let the cat out of the bag

(A) To not allow a cat into the bag
(B) To reveal a secret
(C) To release a cat out of a bag
(D) To buy a cat and a bag

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Education is fundamental for achieving full human potential, developing an equitable and (1)_____ society, and promoting national development. Providing universal (2)_____ quality education is the key to India’s continued ascent, and leadership on the global stage in terms of economic growth, social justice and equality, scientific advancement, national integration, and cultural preservation. Universal high-quality education is the best way (3)_____ for developing and maximising our country's rich talents and resources for the good (4)_____ the individual, the society, the country and the world. India (5)_____ the highest population of young people in the world over the next decade, and our ability to provide high-quality educational opportunities to them will determine the future of our country.

Select the most appropriate option to fill in blank number 1.

(A) such
(B) more
(C) much
(D) just

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Education is fundamental for achieving full human potential, developing an equitable and (1)_____ society, and promoting national development. Providing universal (2)_____ quality education is the key to India's continued ascent, and leadership on the global stage in terms of economic growth, social justice and equality, scientific advancement, national integration, and cultural preservation. Universal high-quality education is the best way (3)_____ for developing and maximising our country's rich talents and resources for the good (4)_____ the individual, the society, the country and the world. India (5)_____ the highest population of young people in the world over the next decade, and our ability to provide high-quality educational opportunities to them will determine the future of our country.

Select the most appropriate option to fill in blank number 2.

(A) access from
(B) access at
(C) access for
(D) access to

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Education is fundamental for achieving full human potential, developing an equitable and (1)_____ society, and promoting national development. Providing universal (2)_____ quality education is the key to India's continued ascent, and leadership on the global stage in terms of economic growth, social justice and equality, scientific advancement, national integration, and cultural preservation. Universal high-quality education is the best way (3)_____ for developing and maximising our country's rich talents and resources for the good (4)_____ the individual, the society, the country and the world. India (5)_____ the highest population of young people in the world over the next decade, and our ability to provide high-quality educational opportunities to them will determine the future of our country.

Select the most appropriate option to fill in blank number 3.

(A) forward
(B) output
(C) outcome
(D) result

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Education is fundamental for achieving full human potential, developing an equitable and (1)_____ society, and promoting national development. Providing universal (2)_____ quality education is the key to India's continued ascent, and leadership on the global stage in terms of economic growth, social justice and equality, scientific advancement, national integration, and cultural preservation. Universal high-quality education is the best way (3)_____ for developing and maximising our country's rich talents and resources for the good (4)_____ the individual, the society, the country and the world. India (5)_____ the highest population of young people in the world over the next decade, and our ability to provide high-quality educational opportunities to them will determine the future of our country.

Select the most appropriate option to fill in blank number 4.

(A) of
(B) for
(C) at
(D) from

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Education is fundamental for achieving full human potential, developing an equitable and (1)_____ society, and promoting national development. Providing universal (2)_____ quality education is the key to India's continued ascent, and leadership on the global stage in terms of economic growth, social justice and equality, scientific advancement, national integration, and cultural preservation. Universal high-quality education is the best way (3)_____ for developing and maximising our country's rich talents and resources for the good (4)_____ the individual, the society, the country and the world. India (5)_____ the highest population of young people in the world over the next decade, and our ability to provide high-quality educational opportunities to them will determine the future of our country.

Select the most appropriate option to fill in blank number 5.

(A) will have

(B) can has

(C) were having

(D) might had

Answer key

Numerical and Mathematical Ability

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(B)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(B)**
Q.8 **(A)**	Q.10 **(A)**	Q.11 **(A)**	Q.13 **(B)**	Q.14 **(B)**	Q.20 **(B)**

Reasoning Ability and Problem Solving

Q.2 **(C)**	Q.3 **(D)**	Q.4 **(B)**	Q.6 **(B)**	Q.7 **(C)**	Q.9 **(D)**
Q.10 **(A)**	Q.11 **(A)**	Q.12 **(C)**	Q.13 **(B)**	Q.14 **(A)**	Q.15 **(D)**
Q.16 **(A)**	Q.18 **(A)**	Q.20 **(D)**			

General Awareness

Q.1 **(C)**	Q.2 **(B)**	Q.3 **(A)**	Q.4 **(A)**	Q.5 **(A)**	Q.6 **(B)**
Q.7 **(C)**	Q.8 **(A)**	Q.9 **(D)**	Q.10 **(D)**	Q.11 **(B)**	Q.12 **(A)**
Q.13 **(D)**	Q.14 **(C)**	Q.15 **(D)**	Q.17 **(C)**	Q.18 **(D)**	Q.19 **(D)**
Q.20 **(C)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(B)**	Q.25 **(C)**

English Language and Comprehension

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(B)**	Q.4 **(C)**	Q.5 **(D)**	Q.6 **(B)**
Q.7 **(B)**	Q.8 **(B)**	Q.9 **(A)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(A)**
Q.13 **(D)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(A)**	Q.17 **(B)**	Q.18 **(A)**
Q.19 **(B)**	Q.20 **(B)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(A)**	Q.24 **(A)**
Q.25 **(A)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (C) 12 minutes

The policeman gains 16 − 15 = 1 km/h on the thief, so closing 200 m = 0.2 km takes 0.2 hours = 12 minutes.

Q.2 · Answer (B) ₹112

(2 × 100 + 3 × 120)/5 = (200 + 360)/5 = ₹112 per kg.

Q.3 · Answer (B) 400

x = 0.8x + 80, so 0.2x = 80 and x = 400.

Q.4 · Answer (B) 72

216 = $2^3 \times 3^3$, 360 = $2^3 \times 3^2 \times 5$ and 504 = $2^3 \times 3^2 \times 7$, so the HCF is $2^3 \times 3^2$ = 72.

Q.5 · Answer (D) 4,725

A pays 7,500 × 8 × 4/100 = ₹2,400. B pays 7,500 × (9 + 10 + 6 + 6)/100 = 7,500 × 31/100 = ₹2,325. Together they pay ₹4,725.

Q.6 · Answer (B) 13.95%

The loss is 21,500 − 18,500 = ₹3,000, and 3,000/21,500 × 100 = 13.95%.

Q.8 · Answer (A) 420

Refrigerator covers = 660 × 7/11 = 420.

Q.10 · Answer (A) 1080

SI = 3,000 × 9 × 4/100 = ₹1,080.

Q.11 · Answer (A) 16 min 30 sec

They meet again at the start after the LCM of 18, 22 and 30 seconds: 2 × 3^2 × 5 × 11 = 990 seconds = 16 min 30 sec.

Q.13 · Answer (B) 178 runs

The 22 innings total 22 × 40 = 880 runs. An average of 46 over 23 innings needs 23 × 46 = 1,058 runs, so he must score 1,058 − 880 = 178.

Q.14 · Answer (B) 750

Sum = 240 × 100/(8 × 4) = ₹750.

Q.20 · Answer (B) 31

(42 × 25 + 28 × 40)/70 = (1,050 + 1,120)/70 = 2,170/70 = 31.

Reasoning Ability and Problem Solving

Q.2 · Answer (C) Only conclusion (I) follows

All jugs are taps, so some taps are jugs (I). The hats that are taps need not be jugs, so II does not follow.

Q.3 · Answer (D) 8

SSC's key gives 8, the only digit in both codes: T is in both words, so its digit must appear in both codes, and none of the other options does. (E is in both words too, so the codes should share two digits; as printed the question is flawed.)

Q.4 · Answer (B) 6

A is the only letter in both words and 6 the only digit in both codes, so A = 6.

Q.6 · Answer (B) 1215

Each term is 3 times the one before: 405 × 3 = 1215.

Q.7 · Answer (C) (3, 14, 3)

The middle number is the product of the other two plus 5: 5 × 7 + 5 = 40 and 12 × 5 + 5 = 65. Likewise 3 × 3 + 5 = 14.

Q.9 · Answer (D) 3

Going right from I the seats run I, L, _, _, J, K, _. M cannot sit next to K, and in the first gap M would put N (fourth to its left) on K's seat; so M takes the second gap, N the seat after K and O the first gap: I, L, O, M, J, K, N. Counting right from O, M, J and K sit before N: three people.

Q.10 · Answer (A) OJ 20

The first letter moves 2 places forward, the second 1 place forward and the number falls by 4: DG 25 → FH 21 and GI 19 → IJ 15. So MI 24 → OJ 20.

Q.11 · Answer (A) 18

After interchanging the signs: 14 – 12 ÷ 2 × 2 + 16 = 14 – 12 + 16 = 18.

Q.12 · Answer (C) K11

Each letter is followed by its place in the alphabet, and the places are the prime numbers 2, 3, 5, 7, 11, 13, 17. The missing term is K11.

Q.13 · Answer (B) ZCUF

Each letter moves 4 places back: UKCT → QGYP and LPBR → HLXN (B wraps round to X). So DGYJ → ZCUF.

Q.14 · Answer (A) Only conclusion I follows

Some potatoes are tomatoes, so some tomatoes are potatoes and I follows. The potatoes that are not tomatoes may be onions, so II does not follow.

Q.15 · Answer (D) (19, 57, 12)

The middle number is the product of the other two divided by 4: 8 × 12 ÷ 4 = 24 and 16 × 5 ÷ 4 = 20. Likewise 19 × 12 ÷ 4 = 57.

Q.16 · Answer (A) 4

B is fourth to A's right and D immediately left of B; G, with two people between it and D counting from G's left, is third to D's right. In the seats left, F must be fourth to E's right and E cannot sit beside B, so going right from A the order is A, E, C, D, B, F, G. Counting right from E, C, D, B and F sit before G: four people.

Q.18 · Answer (A) 5

P is the only letter in both words and 5 the only digit in both codes, so P = 5.

Q.20 · Answer (D) 5

After interchanging the signs: 3 + 2 × 8 ÷ 4 – 2 = 3 + 4 – 2 = 5.

General Awareness

Q.1 · Answer (C) The Padma Awards are given annually.

The Padma awards are announced every year on the eve of Republic Day. Their number is capped at 120 a year (not counting posthumous and foreign awardees), the names are published in the Gazette of India, and the award may not be used as a prefix or suffix to a name.

Q.2 · Answer (B) Plants

Temperature, pressure, humidity, wind, cloudiness and rainfall are the elements of weather; plants are not.

Q.3 · Answer (A) 2006

The 2006 Women's Asia Cup, an ODI tournament, was played in Jaipur in December 2006; India won it.

Q.4 · Answer (A) Jerry Pinto

'Em and the Big Hoom' (2012) is Jerry Pinto's first novel, about a Bombay family living with the mother's mental illness.

Q.5 · Answer (A) the real executive

In India's parliamentary system the President is the nominal executive and the Prime Minister, who heads the Council of Ministers, the real executive. The President is the Supreme Commander of the armed forces.

Q.6 · Answer (B) Amount borrowed from World Bank
Revenue receipts neither create a liability nor reduce assets: taxes such as GST, fees and grants. A loan from the World Bank must be repaid, so it is a capital receipt.

Q.7 · Answer (C) Article 48
Article 48, a Directive Principle, asks the State to organise agriculture and animal husbandry on modern and scientific lines.

Q.8 · Answer (A) 37.45%
By the Carius method, % Br = (80/188) × (0.22/0.25) × 100 = 37.45%, 80 and 188 being the molar masses of Br and AgBr.

Q.9 · Answer (D) 2030
The Union Budget 2023-24 set the National Green Hydrogen Mission a target of 5 MMT of annual production by 2030.

Q.10 · Answer (D) 54.6%
In the 2011 census 54.6% of India's workforce was engaged in agriculture, as cultivators or agricultural labourers.

Q.11 · Answer (B) Bharatanatyam
Alarmel Valli is a Bharatanatyam dancer of the Pandanallur style.

Q.12 · Answer (A) Protein Denaturation
Heat unfolds the proteins of egg white (denaturation), and they bind together into a solid white mass.

Q.13 · Answer (D) Humayun’s tomb
Humayun's tomb in Delhi, built in 1570, was the first garden-tomb on the Indian subcontinent; the tombs of Itimad-ud-Daulah and Safdarjung came later.

Q.14 · Answer (C) Tamil Nadu
The M. A. Chidambaram Stadium is at Chepauk in Chennai, Tamil Nadu.

Q.15 · Answer (D) Arun Sinha
Arun Sinha, an IPS officer who had been an adviser at the National Technical Research Organisation, was appointed its chairman in April 2023.

Q.17 · Answer (C) Assam
In Assam the Nor'westers, violent evening thunderstorms of the hot season, are called 'Bardoisila'; in West Bengal they are 'Kalbaisakhi'.

Q.18 · Answer (D) Gujarat
Rani ki Vav, a stepwell on the Saraswati at Patan in Gujarat, became a UNESCO World Heritage Site in 2014.

Q.19 · Answer (D) 1.64%
The 2011 census put India's average annual exponential growth rate for 2001-2011 at 1.64%, down from 1.97% in 1991-2001.

Q.20 · Answer (C) Pandwani
Usha Barle of Chhattisgarh, a singer of Pandwani, the sung narration of the Mahabharata, received the Padma Shri in 2023.

Q.21 · Answer (D) A, B and C
Balshastri Jambhekar started 'Darpan', the first Marathi newspaper, printed in English and Marathi, on 6 January 1832, and the first Marathi monthly, 'Digdarshan', in 1840; he is called the father of Marathi journalism. All three statements are correct.

Q.22 · Answer (D) Yeast
Yeast ferments sugars into ethanol and carbon dioxide: the gas makes bread dough rise, and the ethanol is the basis of the alcohol industry.

Q.23 · Answer (A) Samhita
A Samhita is the collection of the mantras or hymns of a Veda, such as the Rig Veda Samhita.

Q.24 · Answer (B) 1966
The Indian Forest Service was set up in 1966 under the All India Services Act, 1951, alongside the IAS and the IPS.

Q.25 · Answer (C) Telangana
Bathukamma, a festival of flowers celebrated by women in the autumn, is Telangana's state festival.

English Language and Comprehension

Q.1 · Answer (C) Unquiet
'Tranquil' means calm and peaceful; its opposite is 'unquiet'. 'Serene' and 'composed' are synonyms.

Q.2 · Answer (A) complicated
A problem that took hours to solve was 'complicated'. The blank needs an adjective describing the problem, not the adverbs 'simply' or 'easily'.

Q.3 · Answer (B) scalpture
The correct spelling is 'sculpture'.

Q.4 · Answer (C) Start
'Commence' means to begin: 'start'.

Q.5 · Answer (D) Sarah and Fidha will enter into an agreement to share the resources that are currently accessible.
The simple future 'will enter' puts the agreement in the future; the other options are the present continuous, the past perfect and an ungrammatical passive.

Q.6 · Answer (B) occasion
The correct spelling is 'occasion'.

Q.7 · Answer (B) a
'Passionate' begins with a consonant sound, so it takes the indefinite article 'a'.

Q.8 · Answer (B) even
'Even after' stresses that confession is still good at a point where it might seem too late; 'since', 'though' and 'still' do not fit before 'after' here.

Q.9 · Answer (A) procrastination
The correct spelling is 'procrastination'.

Q.10 · Answer (D) old much
The phrase is 'much too old': 'Ms. Malathi is much too old to join a college now.'

Q.11 · Answer (A) She is the most talented musician in the orchestra.
Comparing one musician with all the others in the orchestra takes the superlative with 'the': 'She is the most talented musician in the orchestra.' 'More talented' would need 'than', and 'talentedest' is not a word. (The question's 'comparative degree' means the degrees of comparison in general.)

Q.12 · Answer (A) smart, tall, Indian
Adjectives of opinion come before those of size or build, and origin comes last: 'a smart, tall, Indian astronaut'.

Q.13 · Answer (D) Type
A 'species' is a kind or 'type', especially of plant or animal.

Q.14 · Answer (A) Proudly
'Submissively' means meekly, yielding to others; its opposite is 'proudly'.

Q.15 · Answer (D) weight; help
One loses 'weight' to avoid health problems, here with the 'help' of a personal trainer.

Q.16 · Answer (A) rest on your laurels
To 'rest on your laurels' is to rely on past success without further effort; a single success could be put down to luck.

Q.17 · Answer (B) a break from social media
One takes 'a break from' something: 'a break from social media'.

Q.18 · Answer (A) Tender
'Callous' means unfeeling and hard-hearted; its opposite is 'tender'.

Q.19 · Answer (B) quietest
Comparing one student with the whole class needs the superlative, and 'quiet' normally forms it with '-est': 'the quietest student of the class'.

Q.20 · Answer (B) To reveal a secret
To 'let the cat out of the bag' is to reveal a secret, usually by accident.

Q.21 · Answer (D) just
'An equitable and just society': 'just' (fair) pairs naturally with 'equitable'.

Q.22 · Answer (D) access to
One provides 'access to' something: 'universal access to quality education'.

Q.23 · Answer (A) forward
'The best way forward' means the best course of action from here on.

Q.24 · Answer (A) of
'For the good of the individual, the society, the country and the world'.

Q.25 · Answer (A) will have
'Over the next decade' points to the future: 'India will have the highest population of young people'.

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